

Sapphire Coding and Usage Guide

Language syntax, standard library namespaces, CLI usage, and verified boundaries.

This manual describes the currently tested implementation in this repository and intentionally avoids claiming production-scale or distributed infrastructure.

It is a local runtime reference for learning, experimentation, scripting, and bounded automation work, not a verified multi-node training or cloud deployment platform.

Language

Sapphire supports bindings, functions, control flow, arrays, maps, structures, lambdas, exceptions, process literals, pipelines, and parallel execution blocks. This is a tree-walk interpreter with local runtime semantics rather than a native-code compiler.

Standard library

The runtime exposes `os`, `fs`, `http`, `ai`, `scheduler`, `ml`, and `agent` namespaces. Several capabilities depend on optional Python packages, hardware, drivers, credentials, or external services.

Execution model

The project includes .sp execution, a REPL, inline eval flows, source inspection, and compiler-driven tooling. These tools help inspect and run scripts without claiming a production-grade distributed platform.

Boundaries

Features are intentionally documented with their current boundaries. Runtime behavior depends on the installed environment and on which optional integrations are available at the time of execution.

Current status summary

Sapphire version 1.0.5 is a Python-based local runtime and language with a working interpreter, bundled tools, and optional integrations. It is appropriate for learning, local automation, and experimental AI workflows. It is not presented as a distributed training stack, a production security-certified platform, or a globally managed service.

Results vary by machine, Python version, OS configuration, library availability, and workload. The documentation here reflects what is actually exercised in this repository and what remains a roadmap item.

Version 1.0.5. Reproduce local measurements with `benchmarks/benchmark_runtime.py`. Values depend on the machine and environment.